Study of the causal effect of work characteristics on job satisfaction and analysis of the usefulness of indexes on its explanation

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Abstract

The aim of this empirical project is to better understand the causal link between job characteristics and job satisfaction of workers. It is also to test the usefulness of job indexes to explain this satisfaction at work. The results show that many characteristics of a job are relevant to explain job satisfaction, and all our results are in line with the literature. Furthermore, we show that job satisfaction indexes have a real explanatory role and can strengthen causal models. Interestingly there exist some differences in the way that the job characteristics impact the job satisfaction among the gender.

1 Introduction

Following the COVID pandemic, many workers began to reflect on their job satisfaction. For example, in the UK, there were 1.3 million job vacancies in December 2021, showing how workers had started questioning their working conditions, expectations, and whether their jobs were meaningful (Allman (2021)). The pandemic highlighted how important it is to work in a supportive and fulfilling environment, while also exposing a lack of flexibility to meet the needs of employees. For companies, dissatisfaction can be costly, as it often leads to higher expenses for hiring, training, and turnover if the worker resigns (Tessema et al. (2022)). It can also harm workers' mental health, even more when employees face long hours or feel undervalued (Van Aerden et al. (2016). Conversely, an improved employee job satisfaction is crucial as it leads to increased productivity at work and impacts the overall performance of the institution (Inuwa (2016). Even if the measure of happiness is highly correlated with each individual's definition of happiness, a common approximate pattern can be recognized (Bond and Lang (2019)). We can extend their conclusions about happiness to job satisfaction and emphasize the importance of developing a clear definition of it. Depending on the authors, different approaches may have been used, which may influence their results. Macdonald and MacIntyre (1997) witnessed two different methods to quantify job satisfaction. The first focused on determining and measuring the main dimensions of job satisfaction, and the second focused on examining the general level of job satisfaction. Thus, examining the different approaches is relevant. One of the most popular definitions of job satisfaction is "a pleasurable or positive emotional state resulting from the appraisal of one's job or job experiences" by Locke (1976).The aim of this study is in the first place to understand what are the causal effects of job characteristics on job satisfaction and then to see how much an index synthesizes these effects. To do so, we use data from the 6th wave of the European Working Conditions Survey (EWCS2015) to compute a multinomial ordinal logistic regression model. We do so to empirically determine whether there exists a causal relation between job characteristics and job satisfaction. To mitigate selection bias in some variables, a propensity score matching is done to clearly establish if there is causality. To answer the second

part of the question, we compute the Job In General (JIG) index and implement it in our model to estimate how much it explains the declared job satisfaction. The EWCS2015 dataset contains a variable job satisfaction scaled from 1 to 4 estimated by the individual. It also contains different variables which correspond to job characteristics. The observation level is the worker. This study contributes to the literature by providing empirical evidence on the causal effect of different job characteristics on the job satisfaction of European workers, mostly in accordance with the literature. In addition, the paper shows how much an index is capable of summarizing the causal effect. In particular, we found that the workers are more likely to be more satisfied when the job implies learning new skills, career opportunities and working in a private company. Moreover, as the salary increases, the chance to be more satisfied at work increases. However, the workers are more likely to be less satisfied if the job implies a mismatch in the number of hours worked, monotonous or complex tasks. As the number of hours worked per week increases, the probability to be less satisfied increases. Interestingly there exist some differences in the way that the job characteristics impact the job satisfaction among the gender. Including, a job satisfaction index in our model, the JIG, allows to explain more precisely the stated job satisfaction. In the first part we take a look at what has been done on the subject for all the variables we study, then we describe the sample, the variables and the methodology we use. In a third part, we present the models that are computed to respond to the question. Finally, we discuss the result and the limitations of this study.

2 Literature Review

Job satisfaction is a concept with no universally agreed-upon definition and is often the subject of debate. One of the most cited definitions (Locke (1976)) states that Job satisfaction can be defined as the feeling that arises when employees perceive that their personal values align with those promoted by their organization.

The effect of **age** on job satisfaction is complicated : on average, workers under 25 years old experience a gap between their desired job and actual jobs, leading to lower job sat-

isfaction but this effect decreases after 25 (O'brien and Dowling (2011)). Aging brings better salaries, rewards, and evolving social roles, which shape expectations and satisfaction (Kalleberg and Loscocco (1983)).

Limited **Career prospects** is one of the main reasons young workers prefer to leave their jobs (Kisto (2016)) because it reduces job satisfaction (Ramhit (2019)). Le Flanchec et al. (2015) support this, noting that promotion prospects shape employees' satisfaction, emphasizing the need for companies to provide reassurance.

Part time is often imposed, and so reduces job satisfaction. This variable needs to be handled with care (Le Flanchec et al. (2015)). The DARES (2024) study reports that 84 percent of the part time workers in France are women. According to some papers, the correlation between part time job and job satisfaction might be positive ((Eberhardt and Shani (1984)) or(Creevy (1995))), negative (Hall and Gordon (1973) or Miller and Terborg (1979)) while some find no correlation (Steffy and Jones (1990) or Thorsteinson (2003)).

Learning measures a worker's opportunity to learn new skills at his job. The correlation may be positive with job satisfaction as it fosters the sentiment of evolution and fulfillment due to the opportunity to learn new skills ((Rowden and Conine Jr (2005)). This variable describes an opportunity thus the case where the worker undergoes training without consent is ruled out. Learning also facilitates career advancement, smoothing the path to promotion and increasing job satisfaction (Ng et al. (2006)). Conversely, the presence of recurrent **monotonous tasks** negatively affects job satisfaction (Melamed et al. (1995)). This may be due to the presence of these tasks often implying a high workload which causes stress and results in a poor work-life balance (Karasek et al. (1987)).

The variable **Private** indicates whether an individual works in the private or public sector. Employment in the public sector is positively correlated with job satisfaction, as public sector roles often involve serving the public interest, which can enhance fulfillment (DeSantis and Durst (1996)). Alternatively, employees in the private sector tend to place greater importance on salary, whereas those in the public sector prioritize job satisfaction (Karl and Sutton (1998)).

The variable **Work Hours** measures weekly hours spent at a paid job. Job satisfaction decreases with both insufficient hours and low wages or excessive hours causing fatigue and work-life imbalance (Wagner (2017)). While higher wages can initially offset long hours, this effect fades beyond a certain threshold, suggesting a U-shaped relationship with an optimal balance.

The variable **Gender** is often preferred over Sex in modern literature. There is no significant direct effect of gender on job satisfaction (Andrade et al. (2019)). However, it influence the impact of other variables : interpersonal relations impact men more while working conditions are more relevant for women (García-Bernal et al. (2005)). Despite these differences, salary and job opportunities are equally important to both genders.

Mismatch Hours refers to a discrepancy between desired and actual working hours, which negatively impacts job satisfaction (Grund and Tilkes (2023)). This effect is stronger for women due to uneven non-work responsibilities. Similarly, underemployment and over-employment further reduce job satisfaction (Pagan (2017)). However, some studies find no significant effect (Lee et al. (2015)), attributing Pagan's findings to other factors linked to part-time work and reduced hours.

The **Salary** is positively correlated with job satisfaction (Lee and Sabharwal (2016)). Private sector workers tend to be more money-driven than those in the public sector (Vandenabeele (2008)). However, according to the adaptation-level theory, individuals quickly adjust to salary increases, causing the initial boost in satisfaction to fade over time (Helson (1947)).

3 Data

The dataset comes from the 6th wave of the European Working Conditions Survey conducted by Ipsos NV for Eurofound across 35 European countries through face-to-face interviews. The study aims to go beyond job quality measurement. Our sample includes 23 variables and 19,904 observations. A key issue is missing salary data : after checking that there is no pattern in the missing values (Refuse to answer, Do not know), we do the hy-

pothesis that values are missing at random. We can then apply hot deck imputation on the variable salary to fill gaps. This imputation is a statistical method that fills the missing values by copying observed values from similar observations of the dataset. To avoid extreme values we use a winsorization which caps the observation to the top and bottom percentile threshold by replacing them with this threshold. The variables used and their distribution are described in the following tables : *Categorical Variables with Proportions*1 *Continuous Variables with Descriptive Statistics*2

4 Models

4.1 First Logit Model

The first model is a cumulative logit model which tries to explore the causality between job satisfaction and the variables chosen. We analyses all coefficients except those for sex and age that are control variables. Let X be the vector of the explanatory variables for the corresponding model. We estimate the following model:

$$\begin{aligned}\log\left(\frac{P(\text{Job_Satisfaction} \leq j \mid X)}{P(\text{Job_Satisfaction} > j \mid X)}\right) &= \tau_j + \beta_1 \cdot \text{Sex} + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Full_Time} + \beta_4 \cdot \text{Private} \\ &+ \beta_5 \cdot \text{Monotonous} + \beta_6 \cdot \text{Complex_Tasks} + \beta_7 \cdot \text{Learning} + \beta_8 \cdot \text{Career} \\ &+ \beta_9 \cdot \text{Work_Hours} + \beta_{10} \cdot \text{Salary_log} + \beta_{11} \cdot \text{Mismatch_hours} + \varepsilon\end{aligned}$$

The results are in **Table 2**. Being a woman increases the probability of being more satisfied at work at the 1 percent level as well as being older , learning new things at work , having growth opportunities in the career and having a higher salary. Working in the private sector also increases the probability of being more satisfied at work but at a 10 percent level. On the contrary, having a monotonous job , handling complex tasks , working more hours and having a mismatch between wanted and actually worked hours decreases the probability of being more satisfied at work at the 1 percent level. Only the variable Full Time displays a non-significant coefficient. McFadden's R^2 of 8.31 percent indicates that the model explains approximately 8.31 percent of the improvement in the likelihood compared to the null model.

4.2 Cross effects model

The second model is also a cumulative logit model with cross-effects added to check for their relevance. We analyses all coefficients except those for sex and age that are control variables. We estimate this model :

$$\log\left(\frac{P(\text{Job_Satisfaction}\leq j|X)}{P(\text{Job_Satisfaction}>j|X)}\right) = \beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{Full_Time} + \beta_3 \cdot \text{Private} + \beta_4 \cdot \text{Monotonous}$$
$$+ \beta_5 \cdot \text{Complex_Tasks} + \beta_6 \cdot \text{Learning} + \sum_{i=2}^{5} \beta_{7_i} \cdot \text{Career}_i + \beta_8 \cdot \text{Work_Hours} + \beta_9 \cdot \log(\text{Salary})$$
$$+\beta_{10}(\text{Age}\times\text{Full_Time})+\beta_{11}(\text{Age}\times\text{Complex_Tasks})+\beta_{12}(\text{Private}\times\text{Work_Hours})+\beta_{13}(\text{Age}\times\text{Sex})$$
$$+\beta_{14}(\text{Full_Time}\times\text{Sex})+\beta_{15}(\text{Private}\times\text{Sex})+\beta_{16}(\text{Monotonous}\times\text{Sex})+\beta_{17}(\text{Complex_Tasks}\times\text{Sex})$$
$$+ \beta_{18}(\text{Learning} \times \text{Sex}) + \sum_{i=2}^{5} \beta_{19_i}(\text{Career}_i \times \text{Sex}) + \beta_{20}(\text{Work_Hours} \times \text{Sex})$$
$$+ \beta_{21}(\log(\text{Salary}) \times \text{Sex}) + \beta_{22}(\text{Sex}) + \varepsilon$$

The results are in **Table 3**. First we can see that some variables become nonsignificant, such as Full Time, Complex Tasks and Work Hours. Then we witness that only a few crossed variables are significant. Having a full time job and growing older decreases the probability of being more satisfied at work at a 5 percent level, as well as being a woman and having an increase in hours worked (level 1 percent), being a woman and growing older (level 10 percent), or being a woman and having an increase in the salary(level 1 percent). These last results may look intriguing, but we can suppose that an increase in salary is often linked to an increase in the workload or responsibility which may not be totally compensated by the salary increase and resulting in a lower satisfaction at work. On the opposite side, being a woman and having a full time job or being a woman and having career growth opportunities seems to increase the probability of being more satisfied at work (respectively at a 1 and 5 percent level). McFadden's R^2 of 8.43 percent indicates that the model explains approximately 8.43 percent of the improvement in the likelihood compared to the null model.

4.3 Matching Model

The Propensity score matching method is used to explore further the causal link of our variables with Job satisfaction and mitigate potential selection bias on Full Time and Mismatch hours variables. Indeed, 72 percent of the part time workers are women in our data which could create a selection bias. On the variable mismatch hours, it is likely that for the workers who will be the most impacted by a mismatch hour it is not possible to consider taking the job so

it will create a potential selection bias. To apply this method, we recode the 4 modalities of job satisfaction variable into a binary, 0 not satisfied and 1 satisfied. The average difference of the probability of being satisfied between control and treatment is estimated via a logit model. The first matching with Full Time as the treatment, take Monotonous as a control variable and Age, Sex, Complex tasks, Private, Work hours, Career and the log of the salary as confounding variables. The matched population is 8,482. We get that a worker going from part time to full time gains 0.032 point of probability of being satisfied which is coherent with the literature and the results from the other models. The full results are in the

Table 4. The second matching with Mismatch hours as the treatment, take Sex, Complex tasks and Learning as control variables and Age, Full time, Private, Work hours, Career, Monotonous and the log of the salary as confounding variables. The matched population is 17,376. We get that a worker going from mismatched hours to matching hours gains 0.094 point of probability of being satisfied which is coherent with the literature and the results of the other models. The full results are in **Table 5**.

4.4 Logit including a Job Index

Measuring job satisfaction is an important area of study for organizational psychology and human resource management. For our study we focused on the Job Descriptive Index (JDI) and the Job In General (JIG). Developed by Smith et al. (1969), the JDI, based on 5 dimensions, evaluates job satisfaction but does not provide an overall score. The JIG, developed by Ironson et al. (1989), overcomes this limitation by providing a global score based on the different score given by the JDI. Castel et al. (2011) highlights that facet measures are correlated with global measures. We decide to use a multiple regression model to estimate the weight of each of the dimensions. We estimate the following model :

$$\text{Job_Index} = \beta_0 + \beta_1 \text{Pay} + \beta_2 \text{Job_Opportunities} + \beta_3 \text{Supervision} + \beta_4 \text{Coworkers} + \beta_5 \text{Work} + \varepsilon$$

The results are in the following table : **Table 5**.

Each variable is a variable of interest and represents a dimension of the JDI. There is no control variable because we are estimating weights. In the first instance, we use a regression without a constant which results in a very high R^2 and a rejection of H0 in the Variance

Inflation Factor (FIV) test showing multicolinearity among variables. The results are such that all dimensions of the index are significant and all contribute positively to overall satisfaction with an R^2 of 27.7 percent. Next, we integrate the job satisfaction index into our first logit model to better assess its link to working conditions. We estimate the following model :

$$\begin{aligned}\log\left(\frac{P(\text{Job_Satisfaction} \leq j \mid X)}{P(\text{Job_Satisfaction} > j \mid X)}\right) = \beta_0 + \beta_1 \text{Sex} + \beta_2 \text{Age} + \beta_3 \text{Full_Time} + \beta_4 \text{Private} + \beta_5 \text{Monotonous} \\ + \beta_6 \text{Complex_Tasks} + \beta_7 \text{Learning} + \sum_{i=2}^{5} \beta_{8_i} \text{Career}_i + \beta_9 \text{Work_Hours} \\ + \quad \beta_{10} \text{Salary_log} + \beta_{11} \text{Mismatch_Hours} + \beta_{12} \text{JDI} + \varepsilon\end{aligned}$$

The results are in this table : **Table 6**. Age and sex are control variables. Several coefficients from the first model show common patterns. Gender and age remain positive and significant, while Full Time remains insignificant, suggesting contract type has no impact on job satisfaction at the 10 percent level. Private becomes non-significant, as do certain Career modalities and Learning , likely because the job satisfaction index captures these effects, confirming its role as a strong predictor. Salary Log has a positive and significant impact, meaning higher salaries increase satisfaction. Conversely, Mismatch Hours and Work Hours have negative effects, showing that schedule misalignment and excessive hours reduce job satisfaction. These findings align with economic literature, where compensation motivates, while poor working conditions cause dissatisfaction.

The inclusion of a job satisfaction index in the model significantly improves the explanation of job satisfaction. Its high coefficient confirms that it plays a fundamental role in structuring satisfaction, by synthesizing the effects of different work dimensions. This integration thus captures a significant portion of the variance that would otherwise be dispersed among several specific explanatory variables. Moreover, the increase in McFadden's R^2 from 7 percent to 17 percent compared to the model without the index indicates a significant improvement in model fit. The inclusion of the JDI enhances the model's ability to explain variations in job satisfaction, reinforcing the idea that satisfaction is a multidimensional construct that can be effectively captured by a global index. This highlights the relevance of using the JDI, particularly for evaluating the influence of various working conditions on employee satisfaction.

5 Conclusion and limits

To conclude, this paper provides, first, a model with multiple variables to explore causality , secondly a model to investigate further the interactions between variables , then a matching method on two variables to deepen the causality with job satisfaction , lastly, the computation of an index and its implementation inside the original model to check its power. Our models perform well, with all variables being significant. However, the variable "Full Time" only becomes significant when interacted with other variables, suggesting that it may require specific conditions within the sample to achieve significance. Most of them have a positive causality with job satisfaction : Sex, Age, Learning, Career, Private and Salary , while Monotonous, Complex Tasks, Work Hours and Mismatch Hours have a negative impact on job satisfaction. Conditioning on being a woman,having career growth opportunities has a positive impact on job satisfaction contrary to the salary, the work hours and the age. That can be explained by the fact that women may give more importance to some job characteristics comparing to men. Including a job index in the model seems to improve the model's capacity to account for variations in job satisfaction, further supporting the notion that satisfaction is a multidimensional concept that can be accurately represented by a comprehensive index. Regarding the matching method, we confirm that it is possible to observe a causal relationship between variables without encountering issues related to selection bias. Implementing the findings of this work could offer benefits for companies. Indeed, enhancing job satisfaction may reduce turnover rates, resulting in lower recruitment and training costs, as well as increased productivity. This allows our study to be relevant for managers to better understand the satisfaction of their employees. It would have been beneficial to have panel data in order to observe the evolution of job satisfaction over time and assess the impact of certain variables throughout the period as well as accessing the precise questions of the JDI and the weights of its dimensions. One can also deplore the lack of instrumental variables in the database which would have helped mitigate endogeneity.

6 Appendix

Table 1: Categorical Variables with Proportions

Variable	**Description**	**Modality (Proportion %)**
Sex	Gender of the individual	0 = Male (49.46%) 1 = Female (50.54%)
Full_Time	Full-time work	0 = No (20.76%) 1 = Yes (79.24%)
Private	Private sector	0 = No (24.57%) 1 = Yes (75.43%)
Monotonous	Job involves monotonous tasks	0 = No (50.80%) 1 = Yes (49.20%)
Complex_Tasks	Job involves complex tasks	0 = No (40.07%) 1 = Yes (59.93%)
Learning	Job involves learning new things	0 = No (30.70%) 1 = Yes (69.30%)
Mismatch_hours	Job implies a mismatch in hours worked	0 = No (57.46%) 1 = Yes (42.54%)
Payment_Individual	Payment based on individual performance	0 = No (85.72%) 1 = Yes (14.28%)
Job_Satisfaction	Job satisfaction	1 = Not at all satisfied (2.72%) 2 = Not very satisfied (12.39%) 3 = Satisfied (59.31%) 4 = Very satisfied (25.58%)

Variable	**Description**	**Modality (Proportion %)**
Career	Prospects for career advancement	1 = Very low (22.26%) 2 = Low (17.07%) 3 = Moderate (22.66%) 4 = High (23.74%) 5 = Very high (14.27%)
Support_Colleagues	Support from colleagues	1 = Never (5.63%) 2 = Rarely (4.66%) 3 = Sometimes (14.60%) 4 = Most of the time (35.28%) 5 = Always (39.83%)
Cooperation_Colleagues	Cooperation among colleagues	1 = Very low (0.89%) 2 = Low (2.02%) 3 = Moderate (7.81%) 4 = High (40.44%) 5 = Very high (48.84%)
Competent_Sup	Competence of the supervisor	1 = Strongly disagree (9.81%) 2 = Tend to disagree (8.79%) 3 = Neither agree nor disagree (16.59%) 4 = Tend to agree (33.11%) 5 = Strongly agree (31.70%)

Variable	**Description**	**Modality (Proportion %)**
Sup_Useful_Comp	Supervisor perceived as useful to the company	1 = Strongly disagree (6.35%) 2 = Tend to disagree (7.31%) 3 = Neither agree nor disagree (15.88%) 4 = Tend to agree (36.49%) 5 = Strongly agree (33.96%)
Support_Manager	Support from manager	1 = Never (7.33%) 2 = Rarely (9.10%) 3 = Sometimes (18.82%) 4 = Most of the time (31.04%) 5 = Always (33.70%)
Useful_Work	Feeling of doing useful work	1 = Very low (1.75%) 2 = Low (2.48%) 3 = Moderate (9.09%) 4 = High (31.42%) 5 = Very high (55.26%)
Appropriately_Paid	Feeling of being appropriately paid	1 = Very low (13.55%) 2 = Low (17.66%) 3 = Moderate (19.00%) 4 = High (32.94%) 5 = Very high (16.84%)

Variable	Description	Modality (Proportion %)
Promo	Change in salary	1 = Decreased a lot (4.33%) 2 = Decreased a little (9.68%) 3 = No change (61.43%) 4 = Increased a little (22.81%) 5 = Increased a lot (1.75%)
Relation	State of relationship with colleagues	1 = Very low (0.97%) 2 = Low (1.66%) 3 = Moderate (6.10%) 4 = High (37.66%) 5 = Very high (53.61%)

Table 2: Continuous Variables with Descriptive Statistics

Variable	Description	Min	Max	Mean	Standard Deviation
Age	Age of the individual	15	87	43.44	11.84
Work_Hours	Number of hours worked per week	1	120	37.58	126.85
Travel_Time	Minutes per day spent traveling to work	1	360	40.29	33.89
Salary	Net monthly earnings (in euros)	251	5001	1392	1009.75

Figure 1: Results of the first model

Estimators for the logit

	Dependent variable:
	Job_Satisfaction
Sex1	0.150***
	(0.030)
Age	0.007***
	(0.001)
Full_Time	-0.037
	(0.045)
Private1	0.057*
	(0.034)
Monotonous1	-0.416***
	(0.030)
Complex_Tasks1	-0.145***
	(0.033)
Learning1	0.185***
	(0.036)
Career2	0.447***
	(0.047)
Career3	0.898***
	(0.045)
Career4	1.238***
	(0.046)
Career5	2.260***
	(0.054)
Work_Hours	-0.010***
	(0.002)
Salary_log	0.230***
	(0.021)
Mismatch_hours	-0.522***
	(0.030)
McFadden R^2	0.0831
Observations	19,904
Log Likelihood	-18,510.970
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 2: Results of the cross effect model

Estimators for the logitwith cross effects regarding gender

	Dependent variable:
	Job_Satisfaction
Age	0.014***
	(0.003)
Full_Time	0.143
	(0.147)
Private1	-0.011
	(0.141)
Monotonous1	-0.371***
	(0.042)
Complex_Tasks1	-0.165
	(0.117)
Learning1	0.195***
	(0.051)
Career2	0.339***
	(0.070)
Career3	0.865***
	(0.064)
Career4	1.282***
	(0.065)
Career5	2.267***
	(0.075)
Work_Hours	-0.006
	(0.004)
Mismatch_hours	-0.553***
	(0.043)
Salary_log	0.302***
	(0.030)
Sex1	1.379***
	(0.320)
Age:Full_Time	-0.007**
	(0.003)
Age:Complex_Tasks1	0.001
	(0.002)
Private1:Work_Hours	0.0002
	(0.003)
Age:Sex1	-0.005*
	(0.003)
Full_Time:Sex1	0.267***
	(0.094)
Private1:Sex1	0.095
	(0.071)
Monotonous1:Sex1	-0.087
	(0.059)
Complex_Tasks1:Sex1	-0.022
	(0.065)
Learning1:Sex1	-0.017
	(0.071)
Career2:Sex1	0.203**
	(0.095)
Career3:Sex1	0.074
	(0.089)
Career4:Sex1	-0.091
	(0.089)
Career5:Sex1	-0.019
	(0.103)
Work_Hours:Sex1	-0.010***
	(0.003)
Mismatch_hours:Sex1	0.063
	(0.059)
Salary_log:Sex1	-0.135***
	(0.041)
R^2 de McFadden	0.0843
Observations	19,904
Log Likelihood	-18,487.000
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 3: Matching for Full Time

Matching for Part Time / Full Time

	Dependent variable:
	Job_Satisfaction_Binary
Full_Time	0.037^{***} (0.021, 0.053)
Constant	0.818^{***} (0.807, 0.830)
Observations	8,186
R^2	0.003
Adjusted R^2	0.002
Residual Std. Error	0.369 (df = 8184)
F Statistic	21.000^{***} (df = 1; 8184)
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 4: Matching for Mismatch Hours

Matching for Mismatch hours

	Dependent variable:
	Job_Satisfaction_Binary
Mismatch_hours	-0.094^{***} (-0.105, -0.083)
Constant	0.881^{***} (0.873, 0.889)
Observations	16,890
R^2	0.016
Adjusted R^2	0.016
Residual Std. Error	0.369 (df = 16888)
F Statistic	272.531^{***} (df = 1; 16888)
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 5: Estimation of 'weights' for each dimension of JDI

Estimation for the 'weights' composing JDI

	Dependent variable:
	as.numeric(Job_Satisfaction)
Pay	0.147^{***}
	(0.003)
Job_Opportunities	0.065^{***}
	(0.003)
Supervision	0.038^{***}
	(0.002)
Coworkers	0.040^{***}
	(0.002)
Work	0.054^{***}
	(0.005)
Constant	0.990^{***}
	(0.033)
Observations	19,904
R^2	0.277
Adjusted R^2	0.277
Residual Std. Error	0.590 (df = 19898)
F Statistic	$1{,}522.624^{***}$ (df = 5; 19898)
Note:	*p<0.1; **p<0.05; ***p<0.01

Figure 6: Logit Model Including a Job Satisfaction Index

Estimators for the LOGIT including the JDI

	Dependent variable:
	Job_Satisfaction
Sex1	0.152***
	(0.031)
Age	0.005***
	(0.001)
Full_Time	-0.048
	(0.046)
Private1	0.019
	(0.035)
Monotonous1	-0.670***
	(0.031)
Complex_Tasks1	-0.167***
	(0.034)
Learning1	0.010
	(0.037)
Career2	0.071
	(0.048)
Career3	0.044
	(0.048)
Career4	-0.041
	(0.052)
Career5	0.279***
	(0.064)
Work_Hours	-0.007***
	(0.002)
Salary_log	0.181***
	(0.021)
Mismatch_hours	-0.360***
	(0.031)
JDI	2.734***
	(0.048)
Observations	19,904
Log Likelihood	-16,676.700
Note:	*p<0.1; **p<0.05; ***p<0.01

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